IFTEKHAR BIN SHOIB

CSE Undergraduate | Computer Vision & AI | Video Surveillance & Edge AI

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate (CGPA 3.79/4.00) focused on computer vision and real-time video surveillance. Building VisionGuard, an AI security platform that turns existing CCTV cameras into a proactive watcher (Grameenphone Future Maker 2026 semi-finalist). Hands-on with Python, OpenCV, YOLO/Ultralytics, PyTorch, dataset preparation, and false-positive analysis; now extending VisionGuard with Frigate NVR, go2rtc, and RTSP/MQTT pipelines.

EDUCATION

B.Sc. in Computer Science & Engineering | American International University-Bangladesh (AIUB) 2023 - Present
CGPA: 3.79/4.00 | Dean's List Honors | Final semester (internship and thesis remaining)

Higher Secondary Certificate (Science) | Uttara High School and College, Dhaka | GPA: 5.00/5.00 2022

Secondary School Certificate (Science) | Feni Govt. Pilot High School, Feni | GPA: 4.50/5.00 2020

PROJECTS

VisionGuard – AI-Powered Video Surveillance Platform | Next.js, TypeScript, YOLOv8n, WebRTC 2026

- Idea: connect to a building's existing cameras through its DVR/NVR (RTSP/ONVIF) and alert the owner only when risk is real, using context — who, what, where and when — instead of plain motion alerts. Designed as a hybrid system: an edge gateway runs fast local checks and sends only key frames to cloud AI.
- Built prototype: person detection with YOLOv8n running in the browser (ONNX Runtime), owner-drawn polygon zones with restricted hours, a rule-based risk score, and Telegram alerts with snapshot and reasons; live phone camera over WebRTC with TURN relay fallback.
- Next step (designed): a gateway using Frigate NVR, go2rtc (RTSP restreaming), and MQTT events to connect VisionGuard to real DVR/NVR cameras.
- Planned: child-safety zones, face recognition (known vs. unknown), weapon detection with a custom YOLO model, suspicious-behaviour detection (loitering, climbing) using tracking, and a multi-signal context engine.

Live: visionguard-ui-three.vercel.app | **GitHub:** github.com/IFTI737/visionguard-ui

Face Verification Door-Lock System | Python, OpenCV, YuNet, SFace, Arduino 2026

- Built a real-time face verification pipeline (YuNet detection + SFace embeddings) using a laptop camera or a phone stream over Wi-Fi; tuned the detection threshold and added blur and landmark checks to reject bad frames.
- Calibrated the match threshold from genuine vs. impostor scores; sends grant/deny to an Arduino servo lock.

Smart Crop Recommendation System for Bangladesh | Python, scikit-learn, FastAPI, ESP32 Fall 2025

- Developed an ML model that recommends crops from soil and weather inputs (N, P, K, pH, rainfall, humidity, temperature), served via a FastAPI REST backend designed for ESP32 sensor data.

Bangla Sentiment Pro | BanglaBERT, XLM-RoBERTa, FastAPI, Hugging Face Spaces 2026

- Fine-tuned BanglaBERT for Bangla sentiment (~91%) and emotion (~70%) and XLM-RoBERTa for Banglish emotion (~62%); deployed with a FastAPI backend on Hugging Face Spaces.

BazarDor – Bangla Market Price Tracker | Next.js, TypeScript, MongoDB, REST APIs 2026

- Built a price-comparison web app across 12 markets with OAuth sign-in, protected routes, and a three-API fallback.

Live: bazar-dor-2-0-ifti.vercel.app | **GitHub:** github.com/IFTI737/bazar-dor-2.0-ifti

RESEARCH EXPERIENCE

Urban Waste Detection with Explainable AI | YOLO26, RT-DETR-L, Faster R-CNN, Ultralytics 2026

- Collected, labeled and organized a ~12,000-image, 9-class waste dataset in Bangladesh (YOLO and COCO formats); trained and compared three detectors — RT-DETR-L performed best (~83.4% mAP@0.50).
- Reviewed detection results with confusion matrices, Grad-CAM++ and Eigen-CAM to find false positives, false negatives, and weak classes (small, distant, occluded objects).

Brinjal Fruit Disease Classification | PyTorch, DenseNet121, MobileNetV2 2026

- Benchmarked 7 CNN backbones and a fusion hybrid on BrinjalFruitX; 91.26% accuracy vs. 80.0% baseline.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, JavaScript, TypeScript, SQL
Computer Vision & AI: OpenCV, YOLO / Ultralytics, PyTorch, TensorFlow, Scikit-learn, ONNX Runtime, Grad-CAM++
Video & Edge AI: WebRTC, live camera streams, ESP32, Arduino; learning Frigate NVR, go2rtc, RTSP / IP cameras, MQTT, Docker
Data: Dataset collection, labeling and validation (YOLO / COCO formats), false-positive / false-negative analysis
Web & APIs: FastAPI, REST APIs, Next.js, React, Node.js, Express.js, Nest.js, MongoDB, PostgreSQL
Tools: Git, GitHub, VS Code, Kaggle (multi-GPU training), Vercel
Spoken Languages: Bengali, English

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

- Semi-finalist, Grameenphone Future Maker 2026 – pitched VisionGuard, an AI-powered CCTV security platform.
- Dean's Award (Dean's List Honors) for a 3.84/4.00 GPA, Spring 2023–2024, AIUB.